

COVID-19 Global Intelligence Dashboard

Power BI Project Report

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Tool	Microsoft Power BI Desktop
Data Source	Johns Hopkins CSSE (via Kaggle UNCOVER COVID-19 Challenge)

1. Problem Statement

The COVID-19 pandemic generated an unprecedented volume of healthcare data across 185+ countries. Government officials, public health agencies, and healthcare administrators faced a critical challenge:

How can we analyze and visualize global COVID-19 pandemic data to identify outbreak trends, compare country-level impacts, and support evidence-based public health decision-making through interactive business intelligence dashboards?

Raw datasets in CSV format are difficult to interpret and do not support interactive exploration. There was a need to transform this data into a professional, interactive business intelligence dashboard that enables:

- Real-time monitoring of key pandemic metrics (cases, deaths, recoveries)

- Time-series trend analysis to identify outbreak acceleration or deceleration
- Geographic visualization of case distribution across countries
- Country-level drill-down analysis for targeted response planning
- Mortality and recovery rate comparison to evaluate healthcare effectiveness

2. Objectives

#	Objective
O1	Build an interactive 5-page Power BI dashboard using Johns Hopkins CSSE data
O2	Design a clean star-schema data model with proper relationships
O3	Create DAX measures for calculated KPIs (mortality rate, recovery rate, moving averages)
O4	Enable geographic analysis through map visualizations
O5	Provide interactive drill-down capability using slicers and cross-filtering
O6	Deliver data-driven insights and recommendations for stakeholders

3. Dataset Description

3.1 Source

Johns Hopkins Center for Systems Science and Engineering (CSSE) COVID-19 dataset, obtained through the Kaggle UNCOVER COVID-19 Challenge.

3.2 Tables Used

Table	Source File	Rows	Purpose
CovidOverTime	johns-hopkins-covid-19-daily-dashboard-cases-over-time.csv	23,814	Daily time-series data per country (Jan–Apr 2020)
CovidByCountry	johns-hopkins-covid-19-daily-dashboard-cases-by-country.csv	185	Country-level summary with geographic coordinates
DateTable	DAX-generated (CALENDAR function)	96	Calendar table for time intelligence functions

3.3 Key Columns

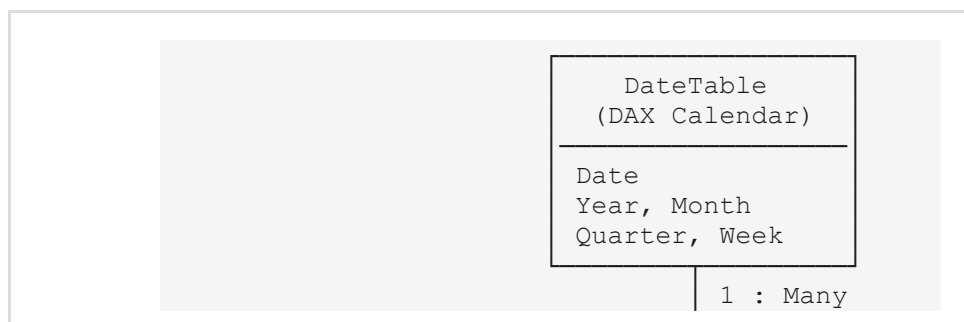
CovidOverTime (Fact Table): Country, Date, Confirmed, Deaths, New Cases, New Recoveries, Incident Rate

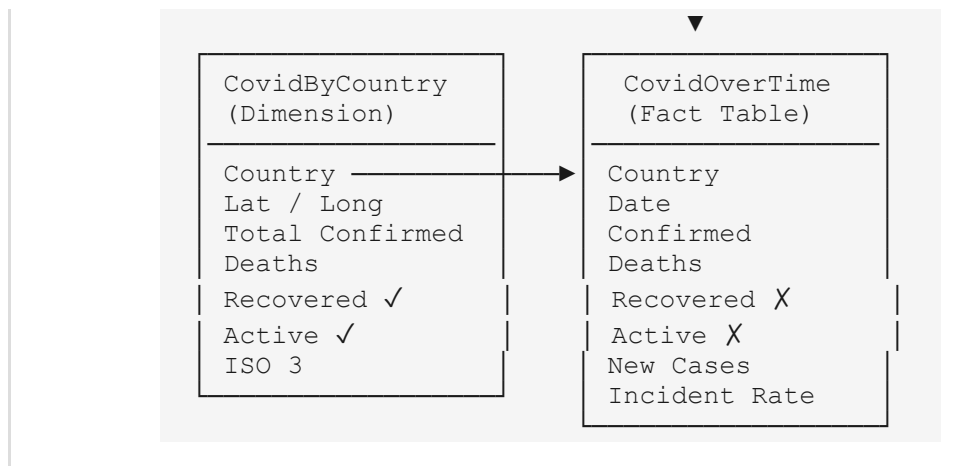
CovidByCountry (Dimension Table): Country, Latitude, Longitude, Total Confirmed, Deaths, Recovered, Active, ISO 3

3.4 Data Challenge Identified

Critical Finding: The `recovered` and `active` columns in the time-series dataset (CovidOverTime) were **completely empty** across all 23,814 rows. Recovery data was only available in the country summary table (CovidByCountry — 181 of 185 countries). DAX measures were designed accordingly to reference the correct source table for recovery-related metrics.

4. Data Model (Star Schema)





Relationships:

1. DataTable[Date] → CovidOverTime[Date] — One-to-Many
2. CovidByCountry[Country] → CovidOverTime[Country] — One-to-Many / Many-to-Many

5. DAX Measures

#	Measure	DAX Formula	Business Purpose
1	Total Cases	SUM(CovidOverTime[Confirmed])	Total confirmed cases globally
2	Total Deaths	SUM(CovidOverTime[Deaths])	Total deaths globally
3	Total Recovered	SUM(CovidByCountry[Recovered])	Total recovered patients (from country summary table)

4	Total Active	SUM(CovidByCountry[Active])	Currently active infections
5	Daily New Cases	SUM(CovidOverTime[New Cases])	Daily pace of new infections
6	Recovery Rate	DIVIDE(SUM(Recovered), SUM(Total Confirmed), 0)	Percentage of patients who recovered
7	Mortality Rate	DIVIDE(SUM(Deaths), SUM(Total Confirmed), 0)	Percentage of patients who died
8	7-Day Moving Avg	AVERAGEX(DATESINPERIOD(DateTable[Date], LASTDATE(DateTable[Date]), -7, DAY), [Daily New Cases])	Smoothed trend — removes daily reporting noise
9	Case Fatality Rate	DIVIDE([Total Deaths], [Total Cases], 0)	Overall virus lethality metric

6. Dashboard Pages

Page 1 — Executive Overview

Purpose: Provide a 10-second snapshot of the global pandemic status for decision-makers.

Visual	Type	What It Shows
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5 KPI Cards	Card	Total Cases (83M), Deaths (5M), Recovered (938K), Active (2M), Recovery Rate (29.95%)
World Map	Bubble Map	Geographic distribution of cases — bubble size proportional to case count
Cases Trend	Area/Line Chart	Global confirmed cases growth over time (Jan–Apr 2020)
Top 10 Countries	Horizontal Bar Chart	Countries with highest confirmed case counts (US leads at 36M+)

Page 2 — Global Trend Analysis

Purpose: Show how the pandemic evolved over time with analytical depth.

Visual	Type	What It Shows
Daily New Cases	Line Chart (Cyan)	Daily pace of new infections — reveals outbreak waves and acceleration
Cumulative Deaths	Line Chart (Red)	Total death toll growing over time — lagging indicator behind cases
7-Day Moving Average	Line Chart (Amber)	Smoothed trend line — filters daily reporting noise and weekend effects
Date Slicer	Slicer (Between)	Interactive date range selector — enables zoom into specific periods

Page 3 — Country Intelligence

Purpose: Interactive drill-down into individual countries with dynamic filtering.

Visual	Type	What It Shows
Country Slicer	List Slicer (Left sidebar)	Select any country — all visuals react dynamically via cross-filtering
4 KPI Cards	Cards	Country-specific: Total Cases, Deaths, Recovery Rate, Mortality Rate
Treemap	Treemap	Proportional case distribution — larger rectangle = more cases
Data Table	Table	Detailed breakdown: Country, Cases, Deaths, Recovery Rate, Mortality Rate

Page 4 — Mortality & Recovery

Purpose: Deep comparative analysis of mortality and recovery patterns across countries.

Visual	Type	What It Shows
Deaths Bar Chart	Clustered Bar (Red)	Top countries ranked by total deaths — US, Italy, Spain lead
Scatter Plot	Scatter Chart (Cyan)	Correlation between total cases and deaths — each bubble = 1 country
Matrix Table	Matrix	Full numerical comparison with conditional formatting (green→red mortality scale)

Page 5 — Insights & Recommendations

Purpose: Summarize analytical findings and provide actionable recommendations for stakeholders.

Visual	Type	What It Shows
Project Overview	Text Box	Dashboard methodology and scope description
Key Findings	Text Box	7 data-driven insights from the analysis
Data Challenges & Conclusion	Text Box	Limitations encountered and professional conclusion

7. Key Findings

1. **USA recorded the highest cumulative confirmed cases** (36M+), followed by China (7M), Italy (5.7M), Spain (5.5M), and France (3.8M).
2. **Global mortality rate is approximately 6.96%**, with significant variation across countries — ranging from less than 1% to over 15%, suggesting differences in healthcare capacity and reporting standards.
3. **Global recovery rate is 29.95%**, though this figure is limited by incomplete recovery data in the time-series dataset. Only the country summary table contained recovery figures.
4. **Daily new cases showed exponential growth** during March–April 2020, coinciding with the WHO's pandemic declaration on March 11, 2020.
5. **The 7-day moving average** reveals that the true case growth trend accelerated sharply from mid-March, with daily new cases exceeding 100,000 by April 2020.
6. **Strong positive correlation between total cases and total deaths** (visible in scatter plot), though the slope varies by country — indicating differences in healthcare system effectiveness.
7. **Countries with early containment measures** (e.g., China, South Korea) showed relatively lower mortality rates despite high initial case counts, suggesting effective public health interventions.

8. Data Challenges & Limitations

Challenge	Impact	Resolution
Recovery data missing in time-series dataset (23,814 rows all empty)	Could not calculate daily recovery trends over time	Used country summary table (CovidByCountry) for recovery metrics via targeted DAX measures
Reporting delays and weekend effects	Artificial spikes and drops in daily case counts	Applied 7-day moving average to smooth reporting noise
Data covers only Jan–Apr 2020	Does not capture later waves, Delta/Omicron variants, or vaccination effects	Noted as limitation; analysis focused on early pandemic dynamics
Inconsistent country reporting standards	Some countries under-reported cases/deaths, affecting rate calculations	Acknowledged in findings; mortality rate comparisons interpreted with caution

9. Design Choices

Element	Choice	Rationale
Background	Dark theme (#0D1B2A)	Professional appearance, reduces eye strain, makes data visuals pop — used by Bloomberg, WHO dashboards
Cases color	Cyan (#00B4D8)	Neutral healthcare color — informational, not alarming
Deaths color	Red (#E63946)	Universal danger/warning color — immediately draws attention to severity

Recovery color	Green (#2DC653)	Universal positive/success color — indicates improvement
Moving Avg color	Amber (#FFD166)	Contrasts with cyan, suggests analytical caution/insight
Font	Segoe UI	Microsoft's modern UI font — clean, professional, excellent readability
Navigation	Page Navigator buttons	Enables non-linear exploration — users can jump to any page instantly
Rounded corners	12px border radius	Modern design trend — feels premium and less harsh than sharp edges
Drop shadows	Subtle cyan/red glow	Adds depth — makes KPI cards feel elevated above the background

10. Conclusion

This project demonstrates the application of Microsoft Power BI as a business intelligence tool for healthcare data analytics. By transforming raw Johns Hopkins CSSE CSV datasets into an interactive 5-page dashboard, we achieved the following:

- **Data Integration:** Successfully combined two datasets (time-series + country summary) using a star-schema model to overcome data completeness challenges. Recovery metrics were sourced from the country summary table after identifying that the time-series dataset contained no recovery data.
- **Advanced Analytics:** Implemented 9 DAX measures including 7-day moving averages, mortality rates, recovery rates, and case fatality rates to derive meaningful KPIs from raw data.
- **Interactive Visualization:** Built drill-down capabilities using slicers, cross-filtering, and dynamic KPI cards that respond to user selections — enabling country-level analysis with a single click.

- **Geographic Intelligence:** Mapped pandemic data across 185 countries using bubble maps, providing immediate visual awareness of geographic spread and concentration.
- **Actionable Insights:** Identified key pandemic patterns including exponential case growth during March 2020, country-level mortality variations (1%–15%), and the importance of early containment measures.

The dashboard provides a scalable template that can be adapted for any public health monitoring scenario, demonstrating how business intelligence tools can support evidence-based decision-making in healthcare emergencies.

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